Band-1 Odd Perfect: Lower Bound / Factor-Chain Constraints Probe (Collapse 1.15)

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Band-1 Odd Perfect Numbers: Published Lower Bound + Classical Constraints (Probe Only)

Honesty stamp. This note records the **published** size lower bound for an odd perfect number (OPN) and certifies a **tiny on-box sample** of classical necessary-condition checks (Euler form, abundancy sketch, ω/Ω/component flags). It does **not** prove that no OPN exists. It does **not** claim a Clay-millennial or ToE result. **GATE-04** is locked out of scope. No factor-chain search to 10^{1500} (or 10^{2200}) was attempted.

Existence of odd perfect numbers remains OPEN. Shrink haystack / ledger only.

Abstract

Published peer-reviewed size lower bound: any odd perfect N satisfies

$$N > 10^{1500}$$

(Ochem–Rao, *Mathematics of Computation* **81** (2012), 1869–1877; DOI 10.1090/S0025-5718-2012-02563-4; author PDF https://www.lirmm.fr/~ochem/opn/opn.pdf).

Author-site computational push: Ochem–Rao state $N > 10^{2200}$ and $\Omega(N) \geq 115$ on https://www.lirmm.fr/~ochem/opn/ — **ledger-noted, not elevated** to published_record (no separate peer-reviewed paper found in 2014–2026 sweep; Integers **24** (2024) still cites 10^{1500}).

Box sample: 11 integers; **7** pass Euler form; **0** odd perfect; all odd samples fail published $\omega \geq 10$, $\Omega \geq 101$, and component $> 10^{62}$ thresholds (as expected for tiny N).

Problem (Band-1 collapse item 1.15)

1. Cite the current **published** lower bound on odd perfect N without invention.
2. Implement classical necessary-condition checks on a **documented sample** (Euler form, abundancy sketch).
3. Keep non-existence proofs, GATE-04, and Clay rhetoric out of the claim surface.

Literature ledger (2024–2026 web check; not re-proved on box)

Year	Bound	Authors	Status
1957	10^{20}	Kanold	historical
1973	10^{36}	Tuckerman	historical
1980s	10^{50}	Hagis	historical
1989	10^{160}	Brent–Cohen	historical

Year	Bound	Authors	Status
1991	10^{300}	Brent, Cohen, te Riele, *Math. Comp.* 57	published
2012	10^{1500}	**Ochem, Rao**, *Math. Comp.* 81	**published_record**
2012+	10^{2200}	Ochem–Rao author site	**author_site_unpublished_extension**
2015	$\omega(N) \geq 10$	Nielsen	published (distinct primes)

Canonical citation for $N > 10^{1500}$. P. Ochem, M. Rao, *Odd perfect numbers are greater than* 10^{1500}, Mathematics of Computation **81** (2012), 1869–1877. DOI: 10.1090/S0025-5718-2012-02563-4. Also: $\Omega(N) \geq 101$; largest component $p^a > 10^{62}$.

Hygiene. Author-site 10^{2200} / $\Omega \geq 115$ are not treated as the published size bound. Zenodo "no OPN" claims are ignored.

Method (box)

1. **Euler form filter.** $N = p^e m^2$ with p prime, $p \nmid m$, $p \equiv e \equiv 1 \pmod 4$ (exactly one odd exponent).
2. **Abundancy.** $\sigma(N)/N$ via multiplicative formula from trial factorization (sample-scale only).
3. **Literature flags.** $\omega \geq 10$, $\Omega \geq 101$, component $> 10^{62}$ — expected fails on tiny sample.
4. **Abundancy product sketch.** Running $\prod p/(p-1)$ over first odd primes — pedagogical end-game tension only; **not** a recreation of the Ochem–Rao proof tree.
5. **Label.** Constraint documentation — **not** a search to 10^{1500}.

Results

Published bound (cited)

Quantity	Value
N_{pub}	$> 10^{1500}$
Source	Ochem–Rao, *Math. Comp.* 81 (2012)
DOI	10.1090/S0025-5718-2012-02563-4
Author-site push 10^{2200} elevated?	**No**

Box sample (certified tonight)

Quantity	Value
Sample size	**11**
Euler-form passes	**7**
Odd perfect in sample	**0**
All odd samples fail $\omega \geq 10/\Omega \geq 101/\text{comp} > 10^{62}$	**True**
Wall time	≈ 0.0002 s

Controls: even perfect 28 (perfect, fails odd/Euler); malformed two-odd-exponent / $p \equiv 3 \pmod 4$ / $e \equiv 3 \pmod 4$ correctly rejected.

Claim surface

Claim	Value
Proves no OPN	**False**
Searched to 10^{1500}	**False**
Elevates 10^{2200} to published	**False**
Documents classical filters on sample	**True**
Cites published $N > 10^{1500}$	**True**

Conclusion

Published lower bound: $N > 10^{1500}$ (Ochem–Rao 2012). Author-site 10^{2200} stays off the published shelf. Box run documents Euler / abundancy / literature-flag filters on a tiny sample; **OPN existence remains OPEN**. GATE-04 locked; no Clay / ToE.

Artifacts

- Code: `experiments/band1/odd_perfect_bounds/opn_constraints_probe.py`
- Results: `results.json`, `literature_ledger.json`, `SUMMARY.txt`